LECTURE PLAN

Institute/Department	UNIVERSITY INSTITUTE OF ENGINEERING (UIE)	Program	Bachelor of Engineering (Information Technology)(IT201)
Master Subject Coordinator Name:	Parampreet Kaur	Master Subject Coordinator E-Code:	E10366
Course Name	Software Testing and Quality Assurance	Course Code	23ITT-307

Lecture	Tutorial	Practical	Self Study	Skilling	TC	TGT	TGP	Studio	Credit	Subject Type
3	0	0	0	0	0	0	0	0	3.00	T

Course Type	Course Category	Mode of Assessment	Mode of Delivery
N.A	Mandatory Non-Graded (MNG)	Theory Examination (ET)	Theory (TH)

Mission of the Department	M1: To provide practical knowledge using state-of-the-art technological support for the experiential learning of our students. M2: To provide industry recommended curriculum and transparent assessment for quality learning experiences. M3: To create global linkages for interdisciplinary collaborative learning and research. M4: To nurture advanced learning platform for research and innovation for students' profound future growth. M5: To inculcate leadership qualities and strong ethical values through value based education.
Vision of the Department	To be recognized as a leading Computer Science and Engineering department through effective teaching practices and excellence in research and innovation for creating competent professionals with ethics, values and entrepreneurial attitude to deliver service to society and to meet the current industry standards at the global level.

Program Educational Objectives(PEOs)	
PEO1	Engage in successful careers in industry, academia, and public service, by applying the acquired knowledge of Science, Mathematics and Engineering, providing technical leadership for their business, profession and community.
PEO2	Establish themselves as entrepreneur, work in research and development organization and pursue higher education.
PEO3	Exhibit commitment and engage in lifelong learning for enhancing their professional and personal capabilities.

Program Specific OutComes(PSOs)	
PSO1	Exhibit attitude for continuous learning and deliver efficient solutions for emerging challenges in the computation domain.
PSO2	Apply standard software engineering principles to develop viable solutions for Information Technology Enabled Services (ITES).

Program OutComes(POs)	
PO1	Disciplinary knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization for the solution of complex engineering problems.
PO2	Complex problem solving: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO3	Critical Thinking
PO4	Creativity
PO5	Communication skills: Communicate effectively on complex engineering activities with the engineering community and with the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO6	Analytical reasoning/thinking: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO7	Research related skills: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.

PO8	Coordinating/Collaborating with others: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO9	Leadership qualities: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO10	Learning how to learn skills: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.
PO11	Digital and technological skills: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
PO12	Multicultural Competence and Inclusive Spirit.
PO13	Value inculcation: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO14	Autonomy, responsibility and accountability: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one’s work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO15	Environmental awareness and action: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and the need for sustainable development.
PO16	Community Engagement & Services.
PO17	Empathy

Text Books

Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Software Testing	Ron Patton	2nd	Pearson Education	2007
2	Software Testing: Principles and Practices	Desikan, S. & Ramesh, G.	2nd	Pearson Education	2006

Reference Books

Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Foundations of Software Testing	Aditya Mathur	1st	Pearson Education	2008
2	Software Quality Assurance – from Theory to Implementation	Daniel Galin	1st	Pearson Education	2009
3	Software Quality Theory and Management	Alan C Gillies	2nd	Cengage Learning	2003

Course OutCome

SrNo	OutCome
CO1	Student will be able to understand the basics of software development life cycle (SDLC) and software testing (ST).
CO2	Student will be able to apply different types of techniques to Implement the test strategies using software.
CO3	Student will be able to analyse the relationship between software modules and different testing methodologies used in industries for software testing.
CO4	Student will be able to monitor test progress of healthcare applications using testing software like JIRA, SOAPUI, POSTMAN etc.
CO5	Student will be able to design test cases to find software bugs.

Lecture Plan Preview-Theory

Unit No	SessionNo	ChapterName	Topic	Text/ Reference Books	Pedagogical Tool**	CO - BT Level Mapping
1	1	Chapter 1.1 :- Fundamental of SDLC (Software Development Life Cycle) and Software Quality	SE basics, SDLC phases, need for process	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT	CO1
1	2	Chapter 1.1 :- Fundamental of SDLC (Software Development Life Cycle) and Software Quality	Waterfall flow, V-Model (V&V), comparison	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT	CO1
1	3	Chapter 1.1 :- Fundamental of SDLC (Software Development Life Cycle) and Software Quality	Agile principles, Scrum, RAD, model comparison	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT	CO1
1	4	Chapter 1.1 :- Fundamental of SDLC (Software Development Life Cycle) and Software Quality	Quality definition, challenges, objectives	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT	CO1
1	5	Chapter 1.1 :- Fundamental of SDLC (Software Development Life Cycle) and Software Quality	Quality Factors & SQA Components	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT	CO1
1	6	Chapter 1.1 :- Fundamental of SDLC (Software Development Life Cycle) and Software Quality	Contract review, dev plan, quality plan	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO1
1	7	Chapter 1.2 Introduction to Software Testing and STCL(Software Testing Life Cycle)	What is testing? Bug definition, errors vs. failures	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO1
1	8	Chapter 1.2 Introduction to Software Testing and STCL(Software Testing Life Cycle)	Tester skills, responsibilities, testing axioms	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO1

1	9	Chapter 1.2 Introduction to Software Testing and STCL(Software Testing Life Cycle)	Test case, suite, defect cycle, severity vs priority	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO1
1	10	Chapter 1.2 Introduction to Software Testing and STCL(Software Testing Life Cycle)	STLC phases: Req ? Plan ? Design ? Execute ? Close	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO2
1	11	Chapter 1.2 Introduction to Software Testing and STCL(Software Testing Life Cycle)	Test policy, test strategy, risk, coverage factors	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO2
1	12	Chapter 1.2 Introduction to Software Testing and STCL(Software Testing Life Cycle)	Test plan, test design spec, test case spec, reports	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO2
1	13	Chapter 1.2 Introduction to Software Testing and STCL(Software Testing Life Cycle)	Structure, entry/exit criteria, summary report	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO2
1	14	Chapter 1.3 :- Test Levels	Testing pyramid, RTM, unit/integration/system/acceptance	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO2
1	15	Chapter 1.3 :- Test Levels	Test Metrics	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO3
1	18	Chapter 2.1 :- Static and Dynamic Testing	Integration, System & Acceptance Testing :- Bottom-up, top-down, system testing, UAT	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO3
2	16	Chapter 2.1 :- Static and Dynamic Testing	Functional vs NFR, examples, importance	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO2
2	17	Chapter 2.1 :- Static and Dynamic Testing	White Box, Black Box, Grey Box :- Concepts, differences, when to use each	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO3

2	19	Chapter 2.1 :- Static and Dynamic Testing	Regression cycles, load/stress testing basics	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO2
2	20	Chapter 2.1 :- Static and Dynamic Testing	Ad-hoc, Exploratory & Website Testing	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO5
2	21	Chapter 2.1 :- Static and Dynamic Testing	Usability & Accessibility Testing	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO2
2	22	Chapter 2.2 :- Software Testing Methodologies	Verification & Validation :-V-model mapping, static vs dynamic testing	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO2
2	23	Chapter 2.2 :- Software Testing Methodologies	White Box Coverage Techniques :-Statement, branch, path, condition, loop coverage	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO3
2	24	Chapter 2.2 :- Software Testing Methodologies	Black Box Design Techniques :- BVA, ECP, state-based testing, decision tables	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO5
2	25	Chapter 2.2 :- Software Testing Methodologies	Cause–Effect Graph & Use Case Testing :- Graphing technique, deriving test cases, scenarios	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5
2	26	Chapter 2.2 :- Software Testing Methodologies	GUI Testing & Test Metrics	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO2

2	27	Chapter 2.3:- Testing Tools	API Testing with SOAPUI & POSTMAN	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO4
2	28	Chapter 2.3:- Testing Tools	Automation: Selenium & Cypress	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO2
2	29	Chapter 2.3:- Testing Tools	Automation	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT	CO2
2	30	Chapter 2.3:- Testing Tools	Test Management Tools :- JIRA overview, IBM RFT basics, reporting & dashboards	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO2
3	31	Chapter 3.1 :- Fundamentals of SQA (Software Quality Assurance)	Definition of quality, evolution of quality, importance	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO1
3	32	Chapter 3.1 :- Fundamentals of SQA (Software Quality Assurance)	Why quality fails, organizational issues, process maturity	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO1

3	33	Chapter 3.1 :- Fundamentals of SQA (Software Quality Assurance)	SQA framework, tasks, audits, checkpoints	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO1
3	34	Chapter 3.1 :- Fundamentals of SQA (Software Quality Assurance)	Software Reviews :- Types: informal, walkthroughs, inspections, peer reviews	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO2
3	35	Chapter 3.1 :- Fundamentals of SQA (Software Quality Assurance)	Formal Technical Reviews (FTR) :- Structure, roles, entry/exit criteria, benefits	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO2
3	36	Chapter 3.1 :- Fundamentals of SQA (Software Quality Assurance)	Formal methods, specifications, model checking, proofs	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO1
3	37	Chapter 3.1 :- Fundamentals of SQA (Software Quality Assurance)	Statistical Quality Assurance:- Sampling, control charts, defect analysis	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO3
3	38	Chapter 3.1 :- Fundamentals of SQA (Software Quality Assurance)	Reliability models	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO3
3	39	Chapter 3.1 :- Fundamentals of SQA (Software Quality Assurance)	ISO 9000 Quality Standards:- ISO 9001 principles, documentation, certification	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO1
3	40	Chapter 3.1 :- Fundamentals of SQA (Software Quality Assurance)	SQA Plan:-Objectives, responsibilities, tasks, deliverables	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO5
3	41	Chapter 3.1 :- Fundamentals of SQA (Software Quality Assurance)	Six Sigma	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO3

3	42	Chapter3.2:- Quality Improvement	Quality Tools: Pareto, Cause–Effect	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO5
3	43	Chapter3.2:- Quality Improvement	Scatter & Run Charts	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO3
3	44	Chapter3.2:- Quality Improvement	Defining Quality Costs, Types of Quality Costs	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO1
3	45	Chapter3.2:- Quality Improvement	Quality Cost Measurement, Utilizing Quality Costs for Decision-making	,T-Software Testing ,T-Software Testing: Principles a,R-Foundations of Software Testin,R-Software Quality Assurance – f,R-Software Quality Theory and Ma	PPT,Video Lecture	CO4

Assessment Model			
Sr No	Exam Name	Max Marks	Weighted Marks
1	External Theory	60	60
2	Assignment/PBL	10	10
3	Attendance Marks	2	2
4	Mid-Semester Test-1	20	10
5	Quiz	4	4
6	Surprise Test	12	4
7	Mid-Semester Test-2	20	10

CO vs PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15
CO1	2	3	2	1	2	3	1	2	2	2	3	2	2	2	2
CO2	2	3	2	1	1	3	1	2	2	2	3	2	2	2	2
CO3	2	3	2	1	1	3	2	2	2	2	3	2	1	2	2
CO4	2	2	2	1	1	2	2	2	2	2	2	2	2	2	2
CO5	2	3	2	2	1	2	2	2	2	2	3	2	2	2	2
Target	2	2.8	2	1.2	1.2	2.6	1.6	2	2	2	2.8	2	1.8	2	2

PO16	PO17	PSO1	PSO2
2	2	3	3
2	2	3	3
2	2	3	3
2	2	3	3
2	2	2	2
2	2	2.8	2.8